



IndianOil

JOURNEY RISK MANAGEMENT (JRM) STUDY

Salem Terminal TO STATE LORRY OWNERS FEDERATION

Objective of the JRM Report

This JRM report is designed to ensure compliance with the Central Motor Vehicle Rules, 1989 (CMVR), AIS 140 standards, and the Road Transport Safety Policy (RTSP). It provides a comprehensive risk assessment for the transportation of hazardous materials along specified routes. By integrating these legal frameworks, the report offers a broad strategy for identifying and mitigating route-specific risks.

Regulatory Compliance

The report complies with the Central Motor Vehicles (Eleventh Amendment) Rules, 2022, mandating safe transportation practices for N2 and N3 category vehicles carrying hazardous materials. These rules require detailed route assessments, especially regarding road conditions, speed limits, and risk areas, to ensure safety compliance.

Risk Management Strategy

This report categorizes transportation routes into high-risk and medium-risk areas, with a focus on factors such as sharp turns, accident-prone regions, and elevation changes. The goal is to provide actionable

recommendations to minimize these risks, including speed regulations, driver warnings for hazardous zones, and the option of alternate routes.

Compliance with the Road Transport Safety Policy (RTSP)

The report integrates RTSP provisions, including mandatory driving hours, rest periods, and nighttime driving restrictions. It ensures that drivers follow official guidelines, such as taking prescribed rest breaks and avoiding dangerous road conditions like poor visibility, heavy crowds, or high-traffic areas during peak hours.

Emergency Preparedness and Response

The report highlights the significance of predetermined emergency stops for refueling, rest, and overnight stays. It includes protocols for safe responses to road hazards, alternative routes, and rerouting processes if roads are closed or severe weather arises. This aligns with the RTSP emphasis on driver safety and rapid emergency response.

Environmental Considerations

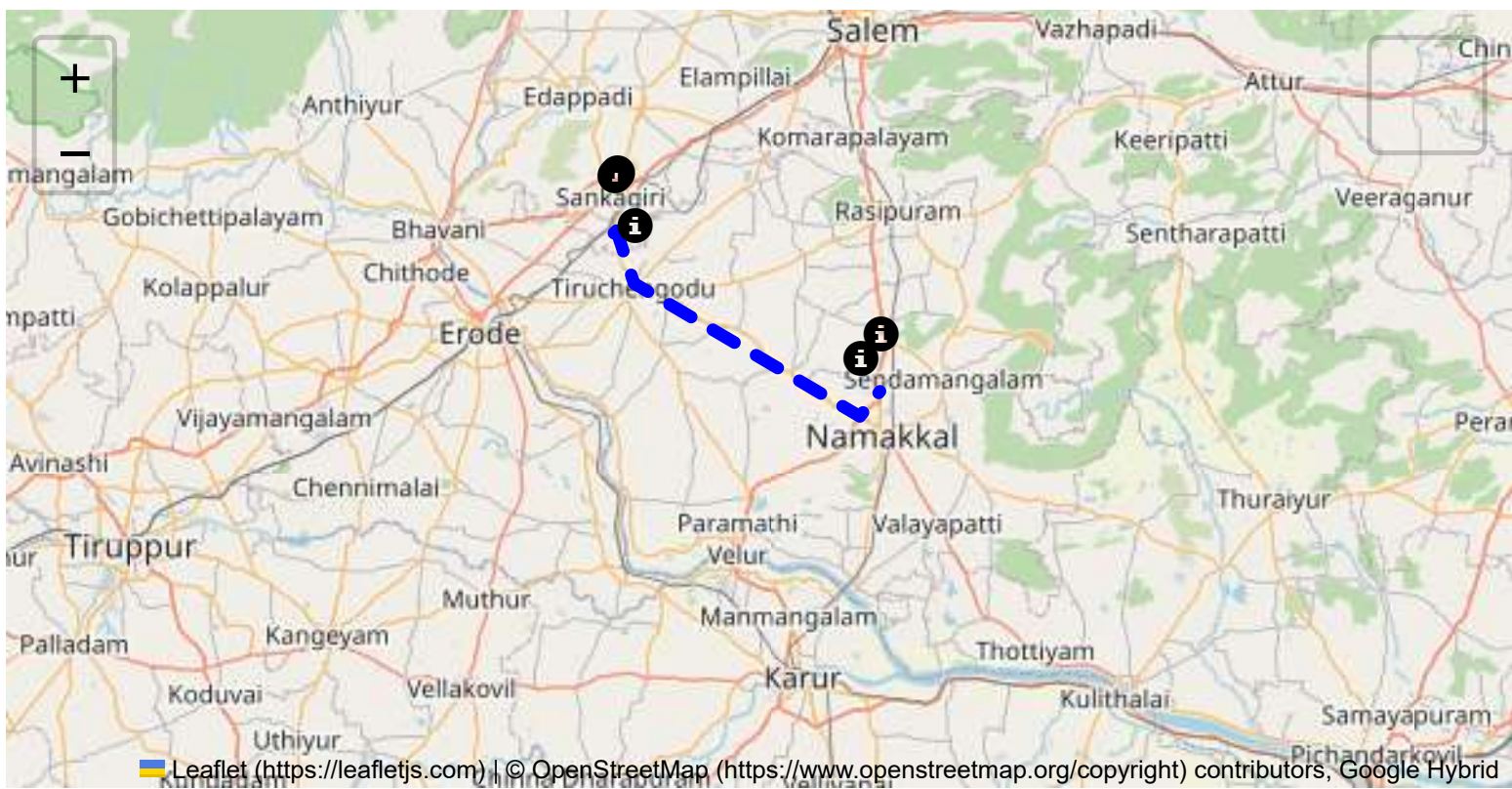
The JRM report addresses environmental risks along the route, ensuring compliance with environmental protection laws in ecologically sensitive zones. It suggests strategies such as identifying areas near water bodies, forests, or populated regions and implementing safety measures to minimize environmental impacts during transport.

Journey Risk Mitigation

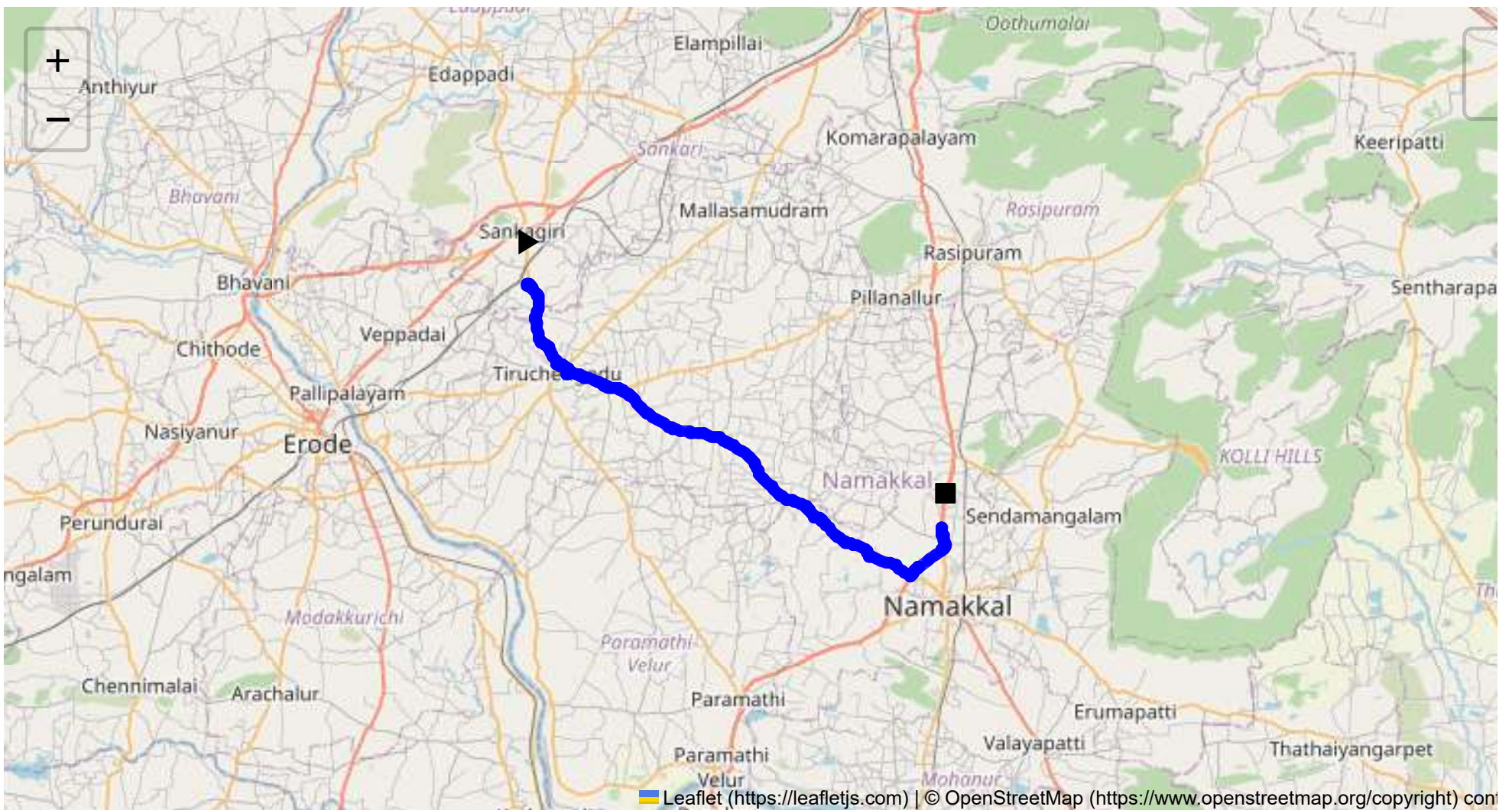
The report includes route-specific risk assessments, detailed journey charts, and defensive driving guidelines for each transport route. Integration with vehicle tracking systems guarantees real-time warnings on hazardous areas, speed limits, and mandatory stops, consistent with RTSP and CMVR safety norms.

Compliance with Government Directives

This report fully adheres to governmental directives regarding hazardous material transportation, implementing mandatory speed limits, nighttime driving restrictions, and comprehensive driver briefings and real-time alerts about route-related risks.



Route Summary:
Total Distance: 46.09 km
Estimated Duration: 1.0 hours
Adjusted Duration (Heavy Vehicle): 1.2 hours
Start: (11.4381, 77.8734)
End: (11.266128, 78.16497)



Welcome to the Journey Risk Management Study

Route Overview

The route from Sangagiri to Pappinayakkanpatti via Kangeyampalayam, Tiruchengode Central Bus Stand, and Ayyampalayam covers approximately 46.09 kilometers. This route travels mainly on regional highways and some state highways, providing connectivity between key towns in Tamil Nadu.

Weather Conditions and Hazards

- **Typical Conditions:** The region generally experiences a tropical climate with a hot and dry summer (March to June) and monsoon rains from June to September.
- **Weather-Related Hazards:** Potential hazards include heavy rainfall during the monsoon season leading to slippery roads and reduced visibility. It is crucial to monitor weather forecasts during these periods.

Traffic Patterns

- **Peak Hours:** Traffic congestion is common in the morning (8:00–10:00 AM) and evening (5:00–7:00 PM) especially around towns such as Tiruchengode.
- **Congestion-Prone Areas:** Expect congestion near Tiruchengode Central Bus Stand and entry/exit points to NH 44.

Road Quality and Infrastructure

- **Assessment:** The primary roads are generally in a reasonable condition, although some sections, particularly the smaller highways, may have potholes or uneven surfaces due to wear and adverse weather.
- **Infrastructure:** Signage and road markings are generally adequate, but care should be taken in less urban areas where they might be less visible.

Alternative Routes for Emergencies

- **Alternative Path 1:** From Sangagiri, one can consider taking the NH 44 directly, bypassing smaller roads. This could offer a safer alternative in some emergency situations but may increase travel time.
- **Alternative Path 2:** Detouring through Rasipuram may also provide a different route trajectory that avoids congestion and poor road conditions.

Local Regulations for Hazardous Material Transport

- Compliance with regulations set by the Tamil Nadu Transport Department is mandatory, including displaying appropriate signage on vehicles and adhering to designated transport corridors where applicable.

Historical Incidents

- **Overview:** There have been occasional incidents involving heavy vehicles mainly due to road conditions and driver fatigue, often exacerbated during the monsoon season. Specific data on hazardous materials is limited.

Environmental Considerations

- **Sensitive Areas:** The route passes close to agricultural zones where runoff from spills could have significant environmental impacts. Maintaining a spill kit on board is advisable.

Communication Coverage

- **Coverage Assessment:** Mobile and data coverage is generally reliable in towns and along major highways. Dead zones are possible in rural stretches, particularly between Kangeyampalayam and Tiruchengode.

Emergency Response Times

- **Estimated Times:** In urban areas such as Tiruchengode, response times can be within 30 minutes. However, rural areas might experience delays of an hour or more due to distance and accessibility.

Overall Risk Assessment Summary

The route presents moderate risks primarily due to weather conditions, road quality, and congestion near urban centers. Adherence to safety and regulatory measures is crucial, especially when transporting hazardous materials. Proper route planning, preparation for adverse weather, and awareness of emergency procedures will mitigate many of these risks. Also, establishing communication protocols and knowing the location of the nearest emergency services along the route can significantly enhance safety for truck drivers.

Risk Assessment - Turns

	Risk Type	Risk Level	Coordinates	Speed Limit
0	Roundabout	High	11.38395, 77.89480	15 KM/Hr
1	U-Turn	High	11.2721854, 78.1639296	10 KM/Hr
2	Turn	High	11.43811, 77.87348	15 KM/Hr
3	Turn	Medium	11.43956, 77.87340	30 KM/Hr
4	Turn	High	11.43968, 77.87345	15 KM/Hr
5	Turn	High	11.44028, 77.87541	15 KM/Hr
6	Turn	Medium	11.38342, 77.89975	30 KM/Hr
7	Turn	High	11.38190, 77.90254	15 KM/Hr
8	Turn	Medium	11.38182, 77.90259	30 KM/Hr
9	Blind Spot	Blind Spot	11.37841, 77.90140	10 KM/Hr
10	Turn	High	11.23961, 78.14223	15 KM/Hr
11	Turn	High	11.24549, 78.14753	15 KM/Hr

	Risk Type	Risk Level	Coordinates	Speed Limit
12	Turn	High	11.24537, 78.14763	15 KM/Hr
13	Blind Spot	Blind Spot	11.27219, 78.16393	10 KM/Hr
14	Turn	High	11.27216, 78.16405	15 KM/Hr

Emergency Locations

	type	name	coordinates	speed_limit	risk_level
0	hospital	Tiruchengode, Goverment Hospital	11.3903328, 77.8920627	30 km/h	Medium
1	hospital	SPM Medical Centre,Tiruchengode	11.3881331, 77.8931963	30 km/h	Medium
5	hospital	Jayaa Hospital	11.3848223, 77.9021886	30 km/h	Medium
8	hospital	Vivekananda Medical Care Hospital	11.364562, 77.9482668	30 km/h	Medium
10	hospital	Manickampalayam, Government Hospital	11.32887, 78.017713	30 km/h	Medium

Crowded Spots

	type	name	coordinates	speed_limit	risk_level
2	school	அரசு ஆண்கள் மேல்நிலைப் பள்ளி	11.3850439, 77.8948279	30 km/h	Medium
3	school	அரசு பெண்கள் மேல்நிலைப் பள்ளி	11.3844247, 77.8949053	30 km/h	Medium
4	marketplace	திருச்செங்கோடு தினசரி காய்கறி சந்தை	11.3833608, 77.8970145	30 km/h	Medium
6	college	Vivekanandha Educational Institutions for Women	11.3596633, 77.9435556	30 km/h	Medium
7	college	Vivekanandha Dental College	11.3643705, 77.9475897	30 km/h	Medium

	type	name	coordinates	speed_limit	risk_level
9	college	Vivekanandha College of engineering for Women	11.3565938, 77.9465342	30 km/h	Medium
11	college	The Surabi College of Polytechnic	11.2660726, 78.0939552	30 km/h	Medium
12	college	CMS College Of Engineering	11.2647508, 78.0969044	30 km/h	Medium
13	school	Porasapalayam Government School	11.2504918, 78.1163024	30 km/h	Medium
14	school	Sri Vinayaga Higher Secondary School	11.2479837, 78.1182908	30 km/h	Medium
15	school	Selvam Higher Secondary School	11.2555236, 78.1671989	30 km/h	Medium
16	school	Anna Nehru School	11.2580962, 78.1672223	30 km/h	Medium

Route Photos of Risky Spots



Risk Type: Roundabout

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.38395, 77.89480



Risk Type: U-Turn

Risk Level: High

Speed Limit: 10 KM/Hr

Coordinates: 11.2721854, 78.1639296



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.44028, 77.87541



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.38342, 77.89975



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.38190, 77.90254



Risk Type: Turn

Risk Level: Medium

Speed Limit: 30 KM/Hr

Coordinates: 11.38182, 77.90259



Risk Type: Blind Spot
Risk Level: Blind Spot
Speed Limit: 10 KM/Hr
Coordinates: 11.37841, 77.90140



Risk Type: Turn
Risk Level: High
Speed Limit: 15 KM/Hr
Coordinates: 11.23961, 78.14223



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.24549, 78.14753



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.24537, 78.14763



Risk Type: Blind Spot

Risk Level: Blind Spot

Speed Limit: 10 KM/Hr

Coordinates: 11.27219, 78.16393



Risk Type: Turn

Risk Level: High

Speed Limit: 15 KM/Hr

Coordinates: 11.27216, 78.16405
